Marketing DAG

JM0190 — 2026 Format: annotated Quarto (`.qmd`) OR Jupyter notebook (`.ipynb`), submitted on Canvas **Weight:** 20% of final grade **Deadline:** [27-05-2026]

Setting

You have been hired as a junior analyst at a mid-sized consumer goods firm. Linda, the marketing director (a bit impatient, sceptical of consultants), three weeks from a board presentation. She wants a defensible answer to a recurring boardroom question: *"How much in additional sales do we get for every additional euro of TV advertising?"* She has heard the term "double machine learning" from a consultant and would like to see whether it changes the answer.

The data team has handed you a dataset (`Marketing_Dataset_for_Individual_Assignment.csv`) and a DAG that the firm's analytics lead drew up last quarter (see Figure 1). Your job is to produce the analysis — and a short, honest summary the director can act on.

The dataset contains some missing values, as is normal in real marketing data. Clean the data as you see fit, and briefly justify your choice in the notebook.

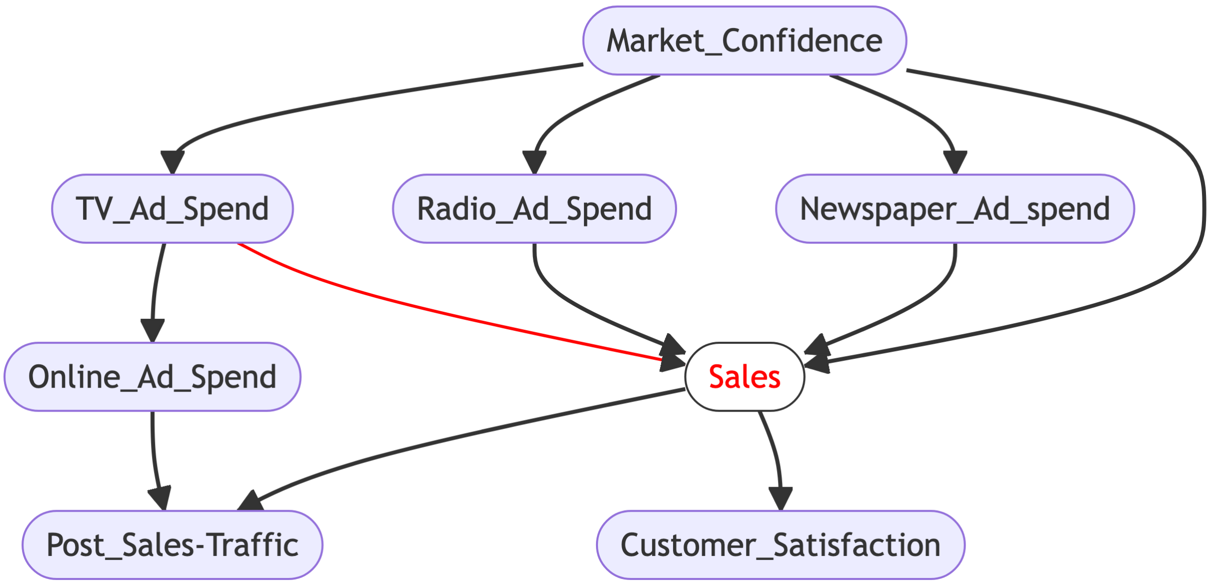

Figure 1. Market_Confidence is an exogenous factor that drives spending on TV, Radio, and Newspaper ads, and also affects Sales directly. TV_Ad_Spend is the treatment of interest; Sales is the outcome. Online_Ad_Spend, Post_Sales_Traffic, and Customer_Satisfaction are all measured *after* sales occur — Online sits downstream of TV, Post_Sales_Traffic downstream of Online, and Customer_Satisfaction downstream of Sales.

What you submit

A single Jupyter notebook (`.ipynb`) OR Quarto (`.qmd`) combining code, output, and short written reflections. Aim for a roughly equal mix of code and prose and for 8–15 rendered pages. Code may be adapted from the lecture slides and notebooks with appropriate citation. Reflections should be specific to *this* dataset, not generic textbook explanations.

You may use AI-based tools as described in the course handbook. You must declare such use clearly at the end of the notebook.

The assignment is in **four parts**, totalling 120 points. Parts 1–3 estimate the average effect of TV ad spend on sales. Part 4 extends the analysis to ask whether the effect varies across the data. Each part builds on the previous ones — start at the top.

Part 1 — Identification (30 points)

Examine the DAG in Figure 1.

1.1 For each variable in the dataset, state its causal role with respect to the effect of `TV_Ad_Spend` on `Sales` (confounder, mediator, collider, descendant of the outcome, or other — be precise). Present this as a short table. *(10 pts)*

1.2 Based on your analysis in 1.1, state which variables you would adjust for in order to identify the causal effect of `TV_Ad_Spend` on `Sales`. Justify your choice in 4–6 sentences. Where relevant, distinguish between variables included for *bias reduction* and variables included for *precision*. *(10 pts)*

1.3 Suppose your colleague suggests, "to be safe, let's just include every variable in the dataset as a control." In 4–8 sentences, explain whether this is a sound idea and why. A clear principled answer is sufficient; stronger answers may also check what happens empirically in this dataset and comment on any gap between the principle and what the data show. *(10 pts)*

Part 2 — Estimation (35 points)

Estimate the causal effect of `TV_Ad_Spend` on `Sales` using **three** approaches:

(i) Naïve OLS — `Sales` regressed on `TV_Ad_Spend` only.
(ii) OLS with the controls you selected in Part 1.
(iii) Double Machine Learning, using the controls you selected in Part 1.

2.1 Report all three point estimates side-by-side. For DML, also report the standard error and the 95% confidence interval. Comment briefly on the precision of the DML estimate: does the interval look like what you would expect, and what is it (and is it not) measuring? *(10 pts)*

2.2 Compare the three estimates. Some will likely agree closely; others will not. In 6–10 sentences, explain what drives each agreement and each disagreement. Be specific: what is each estimator doing, and what does the pattern across the three tell you about the role of your controls and the structure of this dataset? Refer to FWL where helpful. *(15 pts)*

2.3 Of your three estimates, which would you report to the marketing director, and why? *(10 pts)*

Part 3 — Diagnostics and recommendation (35 points)

So far you have trusted the DAG. Now check it.

3.1 Compute the pairwise correlations between all variables in the dataset and present them clearly (a heatmap or a rounded correlation table is fine). Comment briefly on anything you see that seems inconsistent with the DAG in Figure 1. *(10 pts)*

3.2 Pick the one feature of the data you find most striking in light of the DAG. State what the most likely real-world explanation is, and what — if anything — it changes about your analysis in Part 2. You will not be penalised for an interpretation we disagree with, provided it is reasoned and consistent with the data. *(10 pts)*

3.3 Write a short note (max. 200 words) to the marketing director. Include: your best estimate of the effect of TV ad spend on sales, what its uncertainty means in plain language, and one substantive caveat she should know about before acting on it. No equations. No code. The audience does not know what FWL means. *(15 pts)*

Part 4 — Heterogeneity (20 points)

Parts 1–3 estimated a single average effect. That number is correct on average and wrong for everyone individually — campaigns are deployed to specific customers and specific markets, not to averages. The marketing director will eventually want to know not only *whether* TV advertising lifts sales, but *where* it lifts them most.

Extend your Part 2 analysis by fitting a Linear DML model that allows the effect of `TV_Ad_Spend` on `Sales` to vary linearly with one of the variables you adjusted for in Part 1.2. This is the simplest possible CATE model — same residualisation as Part 2(iii), with one treatment-by-covariate interaction term added to the final stage.

4.1 Choose **one** covariate from your Part 1.2 adjustment set to interact with `TV_Ad_Spend`. State your choice and justify it in 2–3 sentences. The justification can appeal to the DAG, to a domain story, or to a pattern you noticed in Part 3 — but not to "every variable might matter." *(5 pts)*

4.2 Fit `econml.dml.LinearDML` with the chosen interaction. Then:

(a) Report the `cate_intercept` and the interaction coefficient, with 95% confidence intervals. Use `dml.summary()`.

(b) State in plain language what the interaction coefficient means for the marketing director. A useful prompt: *"For each one-unit increase in [your chosen covariate], the effect of an additional euro of TV advertising on sales changes by [value]."* Spell it out at least once.

(c) Compute the implied per-euro effect of TV ad spend at the 10th and 90th percentiles of your chosen covariate, and report both numbers. *(10 pts)*

4.3 Rewrite the manager note from 3.3 to incorporate the heterogeneity finding (max. 250 words). Same constraints as 3.3 — no equations, no code, no FWL. The note should state: your best estimate of the average effect, what its uncertainty means in plain language, the heterogeneity finding (where the effect is larger or smaller and by how much), and one substantive caveat. *(5 pts)*

Marking philosophy (for transparency)

This assignment is graded in three tiers, broadly mapping to the points above:

- **Up to 60% — careful reading and execution.** Identify variable roles correctly, run the four estimators, produce honest output, and write coherent prose. Code may be adapted from the lecture notebooks.
- **From 60% to ~85% — judgment.** Defend your choices — including the choice of interaction variable in Part 4. Interpret the gaps between estimates (and the absence of gaps). Read the diagnostics in light of the DAG, not separately from it. Make the manager note land — both versions.
- **The final ~15% — independence.** Notice things the questions don't ask about. Treat the DAG as a hypothesis, not a fact. Make the data-DAG tension productive rather than pretending it isn't there. In Part 4, the strongest answers will treat the heterogeneity finding sceptically — does it survive a different choice of interaction variable, or could it be an artefact of one particular pick?

There is no target number you are trying to hit. A defended estimate is worth more than a "correct" one without justification.